

1)

a) What is the difference between Control Volume analysis and Control System (or Control Mass) analysis? Why is it preferred to use Control Volume analysis for Fluid Flow?

The difference between a Control Mass analysis and a Control Volume analysis is that a Control Mass analysis the mass of the system does not change it is a fixed quantity where as a Control Volume analysis the volume of the system is fixed and the mass can flow through the Control Volume allowing for variability in property characteristics of the fluid. Because of this variability of property characteristics you can determine more detailed information about the fluid interactions with the system. Since a Control mass analysis the mass as it moves through system stays constant the properties of fluid are directly associated to the specific control mass and thus tells us little in terms of fluid interactions, because it would be extremely difficult to track fluid mass elements as they travel through a system.

b) For the integral form of conservation laws, when do we need to setup actual integrations? Under what condition is the time derivative term negligible?

We need to setup integral forms of the conservation laws when we want to analyze the conserved properties as time changes with respect to the systems sphere of interactions (i.e. from time $t = \text{initial interaction}$ to time $t = \text{final interaction}$). Most commonly under steady state conditions is when the time derivative can be neglected as during steady state the rate of change of the conserved properties $= 0$.

c) Write an expression for mass flux through an inlet surface of a rectangular cross-section duct. Fluid velocity vector is $\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}$, density fluid (ρ) is constant, Duct C.S. $A = A$. Assume: the inlet area outward normal vector is $\hat{n} = \hat{i} + 2\hat{j} - \hat{k}$. What is the magnitude of the mass flux if $\rho = 1000 \text{ kg/m}^3$, $A = 1 \text{ m}^2$, $u = 1$, $v = 0$, $w = 3 \text{ m/s}$.

Known: $\dot{m}_i = \rho(\vec{V} \cdot \hat{n})$

Solution: $\int \rho(\vec{V} \cdot \hat{n}) dA \Rightarrow \dot{m}_i = \rho \sum V_i A \quad \vec{V} \cdot \hat{n} = V = u + 2v - w$

$\therefore \dot{m}_i = \rho(u + 2v - w)A = 1000 \frac{\text{kg}}{\text{m}^3} (1 + 2(0) - 3) \text{ m/s} (1 \text{ m}^2) = -2000 \text{ kg/s}$

d) Write expressions for the components of momentum fluxes through the inlet surface based on the velocity and area normal vectors above. Then calculate directions of momentum fluxes ($\hat{i}, \hat{j}, \hat{k}$) respectively.

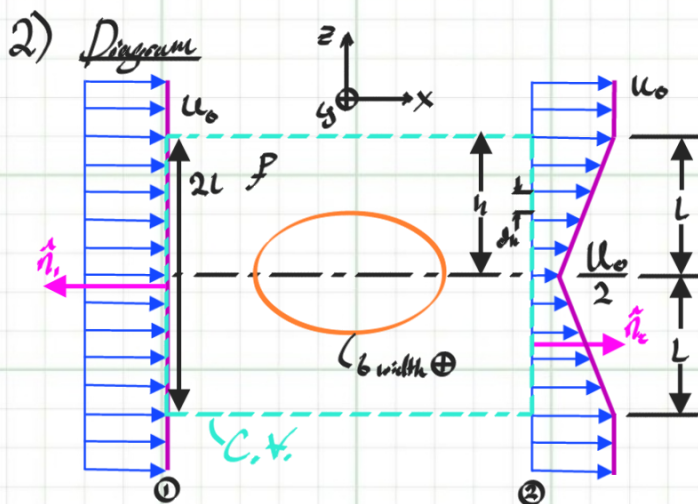
$\int \rho \vec{V}_i \cdot \hat{n}_i dA \Rightarrow \dot{m}_i = \rho(u_i \cdot \hat{n}_i) A = \rho u A = 1000 \text{ kg/s}$

$\int \rho \vec{V}_j \cdot \hat{n}_j dA \Rightarrow \dot{m}_j = \rho(v_j \cdot \hat{n}_j) A = \rho 2v A = 0 \text{ kg/s}$

$\int \rho \vec{V}_k \cdot \hat{n}_k dA \Rightarrow \dot{m}_k = \rho(w_k \cdot \hat{n}_k) A = \rho(-w) A = -3000 \text{ kg/s}$

thus the magnitude and direction for the respective ($\hat{i}, \hat{j}, \hat{k}$) components are

$\dot{m}_i = 1000 \text{ kg/s}$ (+ $\hat{i}$ direction) $\dot{m}_j = 0 \text{ kg/s}$ ($\hat{j}$ direction) $\dot{m}_k = 3000 \text{ kg/s}$ (- $\hat{k}$ direction)



Find

a) Net mass flow rate ($\dot{m}_1$) into C.V. @ state ①

b) Net mass flow rate ($\dot{m}_2$) into C.V. @ State ②

c) Are the two flow rates different? Why? What happens to the difference?

d) height ($2h$, i.e. $\pm h$ about center line) @ state ① that makes $\dot{m}_1 = \dot{m}_2$. Will $h < L$ or $h > L$?

a) $\int d\dot{m}_1 = \int_P \vec{V}_1 \cdot \hat{n}_1 dA$ * Note *

$A_1, 2L \cdot b$ $\vec{V}_1 \cdot \hat{n}_1 = u_{0z} \cdot -\hat{n}_z$

$\Rightarrow \dot{m}_1 = \rho u_0 A_1 = \rho u_0 (2L \cdot b)$ $\Rightarrow V = -u_0$

Assume

- P, u_0, L, b are known Variables

Symmetry in flow is valid about center line

b) $\int d\dot{m}_2 = \int_P \vec{V}_2 \cdot \hat{n}_2 dA = \int_{-h}^h \rho \left(\frac{u_0}{2}\right) b dz$ * Note * $\vec{V}_2 \cdot \hat{n}_2 = \frac{u_0}{2} \hat{i} \cdot \hat{n}_z = V_z = \frac{u_0}{2}$

$\therefore \dot{m}_2 = \rho \frac{u_0}{2} b \int_{-h}^h dz \Rightarrow \rho \frac{u_0}{2} b (z \Big|_{-h}^h) \Rightarrow \rho \frac{u_0}{2} b (h - (-h)) \Rightarrow \rho \frac{u_0}{2} b (2h)$

$\dot{m}_2 = \rho u_0 b (h) = \rho u_0 b L$

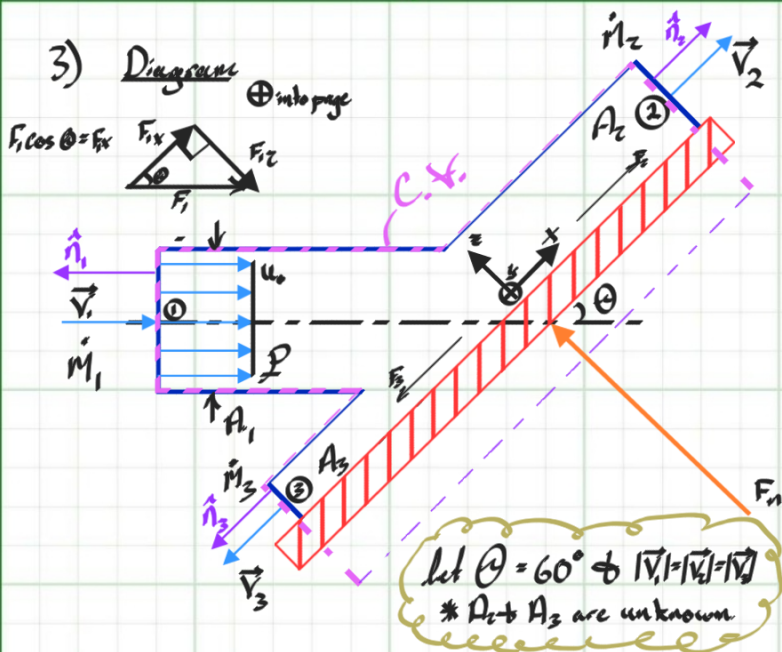
c) the flow rates are indeed different this is because the presence of the object in the middle of the flow impacts the no slip effect of the fluid thus reducing the mass flow as it exits the Control Volume.

as you can see by the magnitudes of $\dot{m}_1$ and $\dot{m}_2$. $\dot{m}_1$ is twice the magnitude of $\dot{m}_2$ which makes sense as the resulting velocity at state 2 is reduced by half.

d) for $|\dot{m}_1| = |\dot{m}_2| \Rightarrow \rho u_0 (2L \cdot b) = \rho u_0 b h$

$\Rightarrow h = 2L$

$\therefore$ the height @ state ① needs to be twice the length but about the center line.



$$\sum F_x = \dot{m}_1 V_1 \cos \theta + \dot{m}_2 V_2 - \dot{m}_3 V_3 + F_{n_x} = 0$$

$$\dot{m}_1 \cos \theta = \dot{m}_3 - \dot{m}_2$$

$$\dot{m}_1 \cos \theta = \dot{m}_3 - \dot{m}_2 \quad \text{eq}^{\#} 1$$

$$\Rightarrow -\rho \dot{Q}_1 \cos \theta = \rho \dot{Q}_3 - \rho \dot{Q}_2$$

$$\Rightarrow \dot{Q}_1 \cos \theta = \dot{Q}_2 - \dot{Q}_3 \quad \text{eq}^{\#} 2$$

$$\sum F_x = \dot{m}_1 V \sin \theta + F_{3z} + F_{2z} + F_n = 0$$

$$\Rightarrow -\dot{m}_1 V \sin \theta = F_n \Rightarrow F_n = \rho \dot{Q}_1 V \sin \theta$$

$$\text{C.O.M.} = \rho V^2 A_1 \sin \theta$$

$$\sum \dot{m} = 0 = \dot{m}_1 + \dot{m}_2 + \dot{m}_3 \Rightarrow -\dot{m}_1 = \dot{m}_2 + \dot{m}_3$$

$$\Rightarrow -(-\rho \dot{Q}_1) = \rho \dot{Q}_2 + \rho \dot{Q}_3 \Rightarrow \dot{Q}_1 = \dot{Q}_2 + \dot{Q}_3 \quad \text{eq}^{\#} 3 \Rightarrow A_1 = A_2 + A_3 \quad \text{eq}^{\#} 4$$

by adding eq[#] 2 & eq[#] 3

$$\dot{Q}_1 + \dot{Q}_1 \cos \theta = \dot{Q}_2 - \dot{Q}_3 + \dot{Q}_2 + \dot{Q}_3$$

$$\Rightarrow \dot{Q}_1 (1 + \cos \theta) = 2 \dot{Q}_2$$

$$\Rightarrow \dot{Q}_2 = \frac{\dot{Q}_1 (1 + \cos \theta)}{2} \therefore \dot{Q}_2 = \frac{V A_1 (1 + \cos \theta)}{2}$$

$$\Rightarrow A_2 = \frac{A_1 (1 + \cos \theta)}{2} \quad \text{eq}^{\#} 5$$

by sub eq[#] 5 into eq[#] 4

$$A_1 = \frac{A_1 (1 + \cos \theta)}{2} + A_3$$

$$\Rightarrow A_1 - \frac{A_1 (1 + \cos \theta)}{2} = A_3$$

$$\Rightarrow \frac{A_1 (2 - (1 + \cos \theta))}{2} = A_3$$

$$\Rightarrow \dot{Q}_3 = \frac{V A_1 (2 - (1 + \cos \theta))}{2}$$

Final

a) Clearly indicate coordinate directions and control volume.

b) Volume flow rate $\dot{Q}_2$ Symbolically

c) Volume flow rate $\dot{Q}_3$ Symbolically

d) Net normal F_n Symbolically

Control Volume marked $C.V. (---)$

Given: $\dot{m} = \rho V A = \rho \dot{Q} \quad V_1 = V_2 = V_3 = V$

$$F = \dot{m} V$$

$$\dot{Q} = \int_A V dA$$

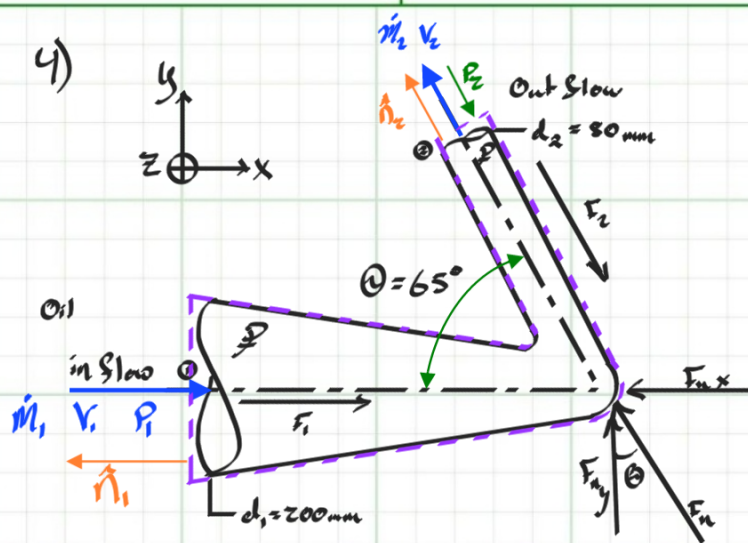
$$\Rightarrow \dot{Q} = \int (\vec{V} \cdot \vec{n}) dA \Rightarrow V A$$

$$\text{For } \vec{n} \text{ into page } \left\{ \begin{array}{l} \dot{m}_1 = -\rho \dot{Q}_1, \text{ since } (\vec{V} \cdot \vec{n}) = -V \\ \dot{m}_2 = \rho \dot{Q}_2 \quad \dot{m}_3 = \rho \dot{Q}_3 \end{array} \right.$$

$$b) \dot{Q}_2 = \frac{3 V A_1}{4} \therefore \cos(60^\circ) = 0.5$$

$$c) \dot{Q}_3 = \frac{V A_1}{4}$$

$$d) F_n = \frac{\rho V^2 A_1 \sqrt{3}}{2} \therefore \sin(60^\circ) = \frac{\sqrt{3}}{2}$$



Find

- Clearly show your Control Volume, Axes define all associated variables.
- What is velocity V_2 Symbolically / numerically
- What is pressure P_2
- Magnitude and direction for the components of the forces required to hold the bend in place then find Net Magnitude and direction.

C.O.M

$$\therefore \dot{m}_1 = \dot{m}_2 \therefore \rho V_1 A_1 = \rho V_2 A_2$$

$$\Rightarrow V_2 = \frac{V_1 A_1}{A_2} \quad A_1 = \frac{\pi d_1^2}{4} \quad A_2 = \frac{\pi d_2^2}{4}$$

$$\therefore V_2 = \frac{(3.5 \text{ m/s}) \left(\frac{\pi (0.2 \text{ m})^2}{4} \right)}{\frac{\pi (0.08 \text{ m})^2}{4}} = 21.875 \text{ m/s}$$

Bernoulli's

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2$$

$$\Rightarrow P_2 = \left(\frac{P_1}{\rho g} + \frac{V_1^2}{2g} - \frac{V_2^2}{2g} \right) \rho g$$

$$\Rightarrow P_2 = P_1 + \frac{(V_1 - V_2)^2 \rho}{2} = 280 \text{ kPa} + \frac{(3.5^2 - 21.875^2) \frac{1}{2} 880 \text{ kg/m}^3}{2}$$

$$= 280 \text{ kPa} - 205.156 \text{ kPa} = 74.843 \text{ kPa}$$

$$P_2 \approx 75 \text{ kPa}$$

d)

$$\sum F_x = \overbrace{\dot{m}_1 V_1 - \dot{m}_2 V_2 \cos(\theta)}^{\text{Volumetric Flow Forces}} + \overbrace{P_1 A_1 - P_2 A_2 \cos(\theta)}^{\text{Pressure Forces}} + F_{1x} = 0$$

$$\Rightarrow -\rho(V_1 A_1) V_1 - \rho(V_1 A_1) V_2 \cos(\theta) - P_1 A_1 - P_2 A_2 \cos(\theta) = -F_{1x}$$

$$\Rightarrow F_{1x} = \rho(V_1 A_1) (V_2 \cos(\theta) + V_1) - P_1 A_1 - P_2 A_2 \cos(\theta)$$

$$\Rightarrow F_{1x} = 7722 \text{ N} \approx 7.7 \text{ kN}$$

$$\sum F_y = -\dot{m}_2 V_2 \sin(\theta) - P_2 A_2 \sin(\theta) + F_{1y} = 0$$

$$\Rightarrow F_{1y} = P_2 A_2 \sin(\theta) + \rho V_2^2 A_2 \sin(\theta)$$

$$F_{1y} = 2259 \text{ N} \approx 2.3 \text{ kN}$$

Known

$$V_1 = 3.5 \text{ m/s} \quad * \text{ Neglect any Energy lost in bends}$$

$$P_1 = 280 \text{ kPa}$$

$$S.G._{oil} = 0.88 = \frac{\rho_{oil}}{\rho_{H_2O}} \Rightarrow \rho_{oil} = 880 \text{ kg/m}^3$$

Assume

- Steady, incompressible flow, No heat loss no frictional loss

- Pipe on horizontal plane i.e. $z=0$

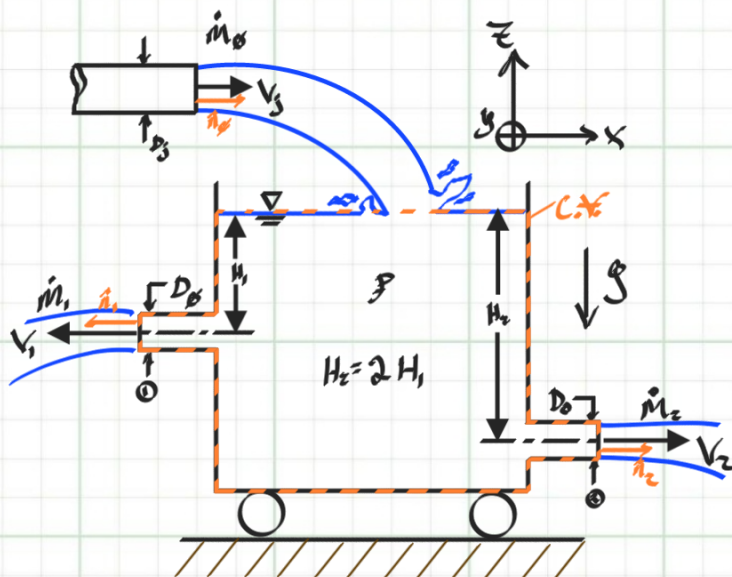
$$\frac{\text{kg m}^2}{\text{s}^2 \text{ m}^3} = \frac{\text{kg m}}{\text{s}^2 \text{ m}^2}$$

$$= \text{N/m}^2 = \text{Pa}$$

$$F_{net} = \sqrt{F_{1x}^2 + F_{1y}^2}$$

$$F_{net} = 8045.9 \text{ N} \approx 8.0 \text{ kN}$$

5)

Find

- V_2 and V_1 symbolically
- Using C.O.M. find relation between V_2 and V_1 , V_2 D, D_0
- Numerical value D_0/D_1 if No horizontal force needed to keep cart stationary

Assume

- inviscid flow
- carts on level surface
- Steady state (filled to point amount of water remains constant)

Bernoulli's

$$\frac{P_0}{\rho g} + \frac{V_0^2}{2g} + z_0 = \frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1, \quad z_1 = -H_1$$

$$0 = \frac{V_1^2}{2g} + (-H_1) \Rightarrow V_1 = \sqrt{2gH_1}$$

$$\text{similarly } 0 = \frac{V_2^2}{2g} + z_2, \quad z_2 = -H_2$$

$$\Rightarrow V_2 = \sqrt{2gH_2} = \sqrt{2g \cdot 2H_1} = \sqrt{4gH_1} = V_1$$

C.O.M.

$$\dot{m}_0 = \dot{m}_1 + \dot{m}_2 \quad \dot{m} = \rho VA$$

$$\Rightarrow V_j A_j = V_1 A_0 + V_2 A_0$$

$$\Rightarrow V_j \pi \frac{d_j^2}{4} = \frac{\pi d_0^2}{4} (V_1 + V_2)$$

$$\Rightarrow V_j = \frac{d_0^2}{d_j^2} (V_1 + V_2)$$

$$\Sigma F_x = -\dot{m}_1 V_1 + \dot{m}_2 V_2 + \dot{m}_0 V_j = 0 \quad F = \dot{m} V$$

$$0 = -\rho V_1 A_0 + \rho V_2 A_0 + \rho V_j A_j$$

$$\Rightarrow V_j^2 A_j = A_0 (V_2^2 - V_1^2)$$

$$\Rightarrow V_j^2 = \frac{d_0^2}{d_j^2} (V_2^2 - V_1^2)$$

$$\left(\frac{d_0}{d_j} \right)^2 (V_1 + V_2)^2 = \frac{d_0^2}{d_j^2} (V_2^2 - V_1^2)$$

$$\left(\frac{d_0}{d_j} \right)^4 (V_1 + V_2)^2 = \left(\frac{d_0}{d_j} \right)^2 (V_2^2 - V_1^2)$$

$$\Rightarrow \left(\frac{d_0}{d_j} \right)^2 = \frac{V_2^2 - V_1^2}{(V_1 + V_2)^2}$$

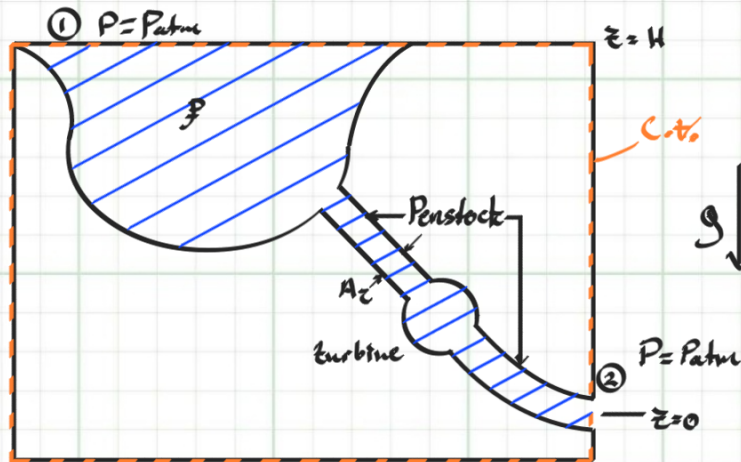
$$\Rightarrow \left(\frac{d_0}{d_j} \right)^2 = \frac{4gH_1 - 2gH_1}{V_1^2 + 2V_1V_2 + V_2^2}$$

$$\Rightarrow \frac{d_0}{d_j} = \sqrt{\frac{2gH_1}{6gH_1 + 2\sqrt{8g^3H_1^3}}}$$

$$\Rightarrow \frac{d_0}{d_j} = \sqrt{\frac{2gH_1}{6gH_1 + 2gH_1\sqrt{8}}} = \sqrt{\frac{2}{6+2\sqrt{8}}}$$

$$\therefore \frac{d_0}{d_j} = 0.4142$$

6)



Finel

- is flow through C.V. Steady or unsteady? How do you know?
- Can one invoke quasi-steady Assumption? What conditions are Assumed for this Assumption.
- Expression for the power generated by turbine symbolically
- Power generated by turbine when $V_z=0$? and when $V_z=\sqrt{2gH}$ what is happening during both states
- Symbolic expression V_z^{crit} at which max power is generated. What is value of Max power.

a) the flow through the C.V. is unsteady
this is because as you drain the reservoir the head pressure at top of penstock will decrease and vice versa as you fill resulting in more work needing to be done by pump as you fill.

B) Yes a quasi-steady state can be invoked because we can assume no friction, No heat loss and because of the duration need to drain and fill can be assumed to be long the variation in instantaneous Velocity changes can be Neglected therefore we can assume Constant Velocities

$$c) \frac{P_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_f + h_t$$

$$h_t = H - \frac{V_z^2}{2g}$$

$$\text{Turbine power} = \rho g Q h_t \Rightarrow \rho Q g \left(H - \frac{V_z^2}{2g} \right)$$